A Comparative Study of Data Generation Strategies for Imbalanced Classification: Synthetic, Simulated, and Augmented Data

Abstract

Class imbalance is a persistent challenge in supervised classification, particularly when minority-class observations are limited. Although oversampling, generative modeling, simulation, and class weighting are frequently used to address this problem, their downstream predictive benefits may depend on the data-generating process and classifier. This study compares baseline learning, class weighting, Synthetic Minority Over-sampling Technique (SMOTE), Conditional Tabular Generative Adversarial Networks (CTGAN), and oracle-based simulation under controlled, reproducible conditions.

Five scenarios represent overlapping class distributions, nonlinear decision boundaries at two noise levels, and distribution shifts involving nuisance or informative variables. Logistic regression and random forest are evaluated using 30 independent evaluation seeds per scenario, with 750 fitting observations, 250 validation observations, and 2,000 test observations per seed. CTGAN training duration was selected separately using development seeds, yielding 10 epochs for both classifiers among the investigated candidates. Average Precision (AP) was designated the primary outcome, with ROC-AUC, F1-score, and balanced accuracy as secondary measures. For each scenario and classifier, four alternatives were compared with the baseline through paired AP differences; two-sided paired t-tests were adjusted using the Holm procedure across 40 comparisons.

The completed experiment generated 3,000 result rows, including default and validation-tuned decision-threshold policies. Twenty-five of the 40 primary comparisons showed a Holm-adjusted statistically significant difference, and all 25 favored the baseline. Across the ten scenario–classifier combinations, the descriptive mean AP differences relative to baseline were -0.0167 for class weighting, -0.0375 for SMOTE, -0.0519 for CTGAN, and -0.0018 for oracle-based simulation. The numbers of significant decreases were six, nine, eight, and two, respectively. No alternative demonstrated a statistically significant AP improvement after multiplicity adjustment. Effects nevertheless varied across scenarios and classifiers, with the oracle-based approach generally remaining closer to baseline than the other alternatives.

These results indicate that generating or reweighting minority-class observations does not guarantee better ranking performance, even when methods are evaluated under a shared experimental framework. They also underline the importance of classifier-specific responses, independent hyperparameter selection, and multiplicity-aware inference. Conclusions are restricted to the simulated settings and implementation choices examined here; the study does not establish that the baseline is universally preferable. The framework provides a reproducible basis for identifying when additional minority-class data may or may not improve imbalanced classification.

Keywords: Synthetic data; simulated data; data augmentation; class imbalance; machine learning.